Data Analytics Capstone Topic Approval Form

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Capstone Project Name: Predicting Cardiovascular Disease with Logistic Regression

Project Topic: Predictive Model for Cardiovascular Disease

☒ **This project does not involve human subjects research and is exempt from WGU IRB review.**

Research Question: Do demographic, lifestyle, and clinical factors significantly impact the likelihood of cardiovascular disease?

Hypothesis:
Null hypothesis- None of the predictor variables are significantly associated with cardiovascular disease.
Alternate Hypothesis- At least one of the predictor variables is significantly associated with an increased likelihood of cardiovascular disease

Context: Cardiovascular disease is the leading cause of death worldwide, with both preventable and genetic risk factors contributing to its diagnoses (CDC). Understanding which demographic, lifestyle, and/or clinical variables significantly influence cardiovascular disease can help in identifying at-risk populations and improve preventative measures. By using logistic regression, this project will determine the significance and magnitude of associations between the predictors and the risk of cardiovascular disease.

Data: The project will be using the publicly available "Cardiovascular Disease Dataset from Kaggle (Welkins *n.d.)*

The dataset has approximately 63.2K records and 16 variables, including:

- ID (categorical, identifier only)
- Age in days (continuous)
- Age in years (continuous)
- Gender (categorical)
- Height (continuous)
- weight (continuous)
- Systolic blood pressure / ap_hi (continuous)
- Diastolic blood pressure / ap_lo (continuous)
- Cholesterol (categorical, coded 1–3)
- Glucose (categorical, coded 1–3), smoking status (categorical, binary 0/1)
- Alco (alcohol intake, categorical, binary 0/1)
- Active (physical activity, categorical, binary 0/1)
- BMI (continuous, derived from weight and height)
- BP_category (categorical, based on ap_hi and ap_lo)
- BP_category encoded (categorical)
- Cardio (categorical, binary 0/1, target variable)

Part of the data originates from the University of California Irvine Machine Learning Repository. According to the website, it is licensed under Creative Commons Attribution 4.0 International, meaning it "…allows for the sharing and adaptation of the datasets for any purpose, provided that the appropriate credit is given." (Janosi, et.al 1989). Another part of the dataset originates from another Kaggle dataset "Heart Disease Dataset," which is licensed under Public Domain, meaning there is no copyright and rights to the work have been waived (Yasser *n.d.*). The "Cardiovascular Disease Dataset" gives appropriate credit to the UCI dataset at the beginning of the description section, and the "Heart Disease Dataset" is free to use, therefore it is permissible to use.

Data Gathering: No original data collection is required. The dataset will be downloaded directly from Kaggle, cleaned, and preprocessed for analysis. Missing values and null values will be detected and handled through imputation. Outliers, if any, will be left as is to keep data integrity. Categorical variables will be encoded using dummy variables.

Data Analytics Tools and Techniques: Python will be the primary programming language for this analysis, supported by libraries such as pandas, numPy, scikit-learn, statsmodels, matplotlib, and seaborn. Logistic regression will be used to model the relationship between the multiple predictor variables and the binary target variable (presence or absence of cardiovascular disease).

Justification of Tools/Techniques: Logistic regression is the appropriate for this project because the target variable, cardiovascular disease, is binary (present (1) /absent (0)). This method is used in healthcare research to predict the likelihood of disease but to provide interpretable results through odds ratio, which indicates whether a factor increases or decreases risk. Logistic regression tests the statistical significance for each predictor, which directly addresses the research question of whether demographic, lifestyle, or clinical factors influence cardiovascular disease. It also allows multiple predictors to be analyzed at once in a single model for ease.

Project Outcomes: The anticipated outcomes of this project include the identification of factors that significantly impact cardiovascular disease. A logistic regression model will quantify the strength and direction of the factors using odds ratio. Backward elimination will be used to find the most statistically significant variables, using VIF and p-value. The model will be evaluated using accuracy score and confusion matrix to assess how well the model predicts cardiovascular disease. Visualizations like histograms and scatterplots will be included to show variable distribution and potential outliers present. A written report will also be included, summarizing the methodology, results, and implications of the model, along with recommended course of action.

Projected Project End Date: 08/26/2025

Sources:

Centers for Disease Control and Prevention. (2024, October 24). *Heart disease facts and statistics.* U.S. Department of Health & Human Services. https://www.cdc.gov/heart-disease/data-research/facts-stats/index.html

Janosi, A., Steinbrunn, W., Pfisterer, M., & Detrano, R. (1989). Heart Disease [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C52P4X.

Welkins, C. (*n.d.*). *Cardiovascular disease dataset.* Kaggle. https://www.kaggle.com/datasets/colewelkins/cardiovascular-disease

YasserH. (*n.d.*). *Heart disease dataset.* Kaggle. https://www.kaggle.com/datasets/yasserh/heart-disease-dataset

Course Instructor Signature/Date:

☐ The research is exempt from an IRB Review.

☐ An IRB approval is in place (provide proof in appendix B).

Course Instructor’s Approval Status: Approved

Date: Click here to enter a date.

Reviewed by:

Comments: Click here to enter text.